

CS104: Software Engineering & System Architecture

Chapter 1: Software Development Lifecycle (SDLC)

Software Engineering applies systematic principles to the design, development, testing, and maintenance of software systems.

Waterfall Model is a linear sequential SDLC model with distinct phases: Requirements, Design, Implementation, Testing, Deployment, Maintenance.

Agile Methodology emphasizes iterative development, flexible collaboration, customer involvement, and rapid response to change.

Scrum is an Agile framework structured around fixed-length Sprints (typically 2-4 weeks), daily standup meetings, Sprint Planning, Sprint Review, and Retrospectives.

User stories describe features from the perspective of an end-user and include acceptance criteria.

Chapter 2: Software Architecture Styles

Software architecture defines the high-level structure of software systems and component relationships.

Monolithic Architecture builds the entire application as a single unified deployable unit. Simple to start, but difficult to scale independently.

Microservices Architecture decomposes an application into independent, loosely coupled, independently deployable services communicating via HTTP/REST or gRPC.

Layered (N-Tier) Architecture organizes code into distinct logical layers: Presentation Layer, Business Logic Layer, Data Access Layer, and Database Layer.

Event-Driven Architecture decouples producers and consumers using asynchronous event streams via message brokers like Apache Kafka or RabbitMQ.

Chapter 3: RESTful API Design & Principles

Representational State Transfer (REST) is an architectural style for designing networked applications over HTTP.

REST Architectural Constraints: Client-Server separation, Statelessness, Cacheability, Uniform Interface, Layered System, and Code on Demand.

Standard HTTP Methods: GET (retrieve resource), POST (create resource), PUT (replace resource), PATCH (partial modification), DELETE (remove resource).

HTTP Status Codes: 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 422 Unprocessable Entity, 500 Internal Server Error.

OpenAPI/Swagger provides standard specification for documenting RESTful API endpoints and schemas.

Chapter 4: Quality Assurance & CI/CD Pipelines

Software testing ensures software correctness, quality, and performance.

Testing Levels: Unit Testing (tests individual functions/classes), Integration Testing (tests component interactions), System Testing (tests entire integrated system), End-to-End Testing (simulates real user flows).

SOLID Principles for object-oriented design: Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, and Dependency Inversion.

Continuous Integration (CI) automatically builds and runs test suites whenever code changes are committed.

Continuous Deployment (CD) automatically deploys validated code changes to production environments.